

Histopathology images are one of the most common tools used in cancer research and diagnosis. H&E slides are already part of standard pathology workflows. They are relatively simple to obtain, but do not directly show the molecular and protein features that matter most for prognosis and treatment. Getting that kind of information means using other methods like multiplex protein imaging or molecular testing, which are expensive, slow, and hard to scale up. Researchers ask whether machine learning can pull deeper biological information out of these routine tissue images.

This review looks at two recent papers that tackle that question in related but somewhat different ways. Andani et al., in "Histopathology-based protein multiplex generation using deep learning," introduce a model called HistoPlexer that predicts spatial protein maps directly from H&E-stained tissue images, which aims to provide a faster and more scalable alternative to direct multiplex protein imaging. El Nahhas et al., in "Regression-based Deep-Learning Predicts Molecular Biomarkers from pathology slides," examine whether a regression-based deep learning approach can outperform classification-based methods that are more common in this field. The outputs are different, but both papers are built on the same core idea. Routine pathology images probably contain more biological information than anyone can see by eye, and deep learning methods should be able to recover most of it.

The methods in the two papers show different machine learning goals. HistoPlexer was trained on paired H&E images and imaging mass cytometry data collected from metastatic melanoma samples. It predicts 11 protein markers simultaneously, producing spatial protein multiplex maps across a tissue section. It shows a supervised deep learning model doing image-to-image prediction, which is built on a conditional generative adversarial network. One part of the model generates protein maps from H&E input, and another part evaluates whether those maps look realistic compared to actual protein data. Training both parts together pushes the model toward more accurate outputs. That design makes sense because the goal is not predicting a single number. Rather, it generates a full spatial map that preserves real tissue-level patterns.

The El Nahhas et al. model takes whole-slide pathology images from patients across multiple cancer types and predicts continuous biomarker values. That distinction matters because many biomarkers do not fall neatly into categories, more so, they exist on a spectrum. Converting a continuous measurement into a binary label removes real information, and the authors argue that this is part of why classification tends to underperform for this kind of problem. The model combines deep learning-based feature extraction with attention-based multiple-instance learning, allowing it to identify the most relevant regions of a slide rather than treating the entire image equally. Since pathology slides are enormous, this becomes an important design choice.

Both studies achieved strong results, though they measured success differently because the problems were not the same. HistoPlexer was evaluated by comparing its predicted protein maps

against real multiplex imaging data. It outperformed baseline methods on image similarity metrics. The authors then went on to test whether those generated maps were useful downstream. They found that HistoPlexer outputs improved tasks like survival prediction and immune subtype classification. These are meaningful findings because they suggest the maps were not just visually convincing, but biologically informative too.

El Nahhas et al. focused more on the direct comparison between regression and classification. Their regression model predicted biomarkers more accurately and showed stronger clinical relevance when tested on external validation datasets. The predictions also turned out to be more informative for survival outcomes, which supports the paper's main argument that regression is a better fit when the target variable is naturally continuous. Forcing a complex biological measurement into categories seems to cost something, and this paper makes a decent case for why.

Each paper has its own strengths worth noting. HistoPlexer is impressive because it goes beyond image classification and tries to reconstruct spatial protein information from standard histology slides, which is useful from a biology standpoint. Especially since protein expression patterns matter a lot in cancer. Predicting multiple markers at once rather than one at a time also seems like an advantage because it may help preserve real relationships between proteins. That said, the model needs paired training data from specialized assays. These are hard and expensive to collect, and the outputs are still predictions rather than actual measurements, so there are some limitations there.

El Nahhas et al. stand out more on the machine learning side. They directly compare regression against classification, and use multi-cancer datasets with external validation to come to more clean conclusions. One limitation is that continuous biomarker labels can be noisy, which makes regression harder, but the paper argues that the benefits of preserving that continuous information are worth it, and the results agree.

Even though the papers are related, they push the field forward in different directions. HistoPlexer is more focused on generating virtual spatial proteomics from tissue slides, which moves computational pathology toward richer molecular interpretation. El Nahhas et al. are more focused on how biomarker prediction should be framed as a problem, and they make a convincing case for regression over classification when needed. Together, these studies suggest that deep learning in computational pathology is moving beyond basic image labeling toward models that can recover real biological detail from slides already used in clinical workflows. If that holds up, it could make pathology more informative and more useful for precision medicine without requiring entirely new infrastructure to do it.

